

1. Administrative & Study Identification

**Full Study Title:** A Study of MK-8527 to Prevent Human Immunodeficiency Virus Type 1 (HIV-1) (MK-8527-010) **Official Title:** A Phase 3, Randomized, Active-Controlled, Double-Blind Clinical Study to Evaluate the Efficacy and Safety of MK-8527 Oral Once-Monthly as HIV-1 Preexposure Prophylaxis in Women **Short Title/Acronym:** EXPrESSIVE-10 **Protocol Number:** MK-8527-010 **NCT Number:** NCT07071623 **Sponsor/Company Name:** Merck Sharp & Dohme LLC / Merck **Investigational Product:** MK-8527 (Nucleoside reverse transcriptase translocation inhibitor / NRTTI) **Phase of Development:** Phase 3 **Trial Status:** RECRUITING (Currently recruiting participants) **Disease Areas / Conditions:** Human Immunodeficiency Virus (HIV), HIV Pre-Exposure Prophylaxis **Medical Monitor:** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate if taking MK-8527 once a month works to prevent HIV-1 infection better than a standard (usual) pre-exposure prophylaxis (PrEP) taken once a day. **Secondary Objectives:** To evaluate the safety of MK-8527 and if people tolerate it.

**2.2 Background & Rationale** Current HIV PrEP options primarily include daily oral pills or long-acting injectables, which can pose real-world challenges related to adherence and stigma. MK-8527 is a novel investigational oral nucleoside reverse transcriptase translocation inhibitor (NRTTI) that offers sustained drug levels, allowing for once-monthly dosing. This study aims to provide a more convenient prevention paradigm for populations at high risk of HIV acquisition, specifically focusing on women in sub-Saharan Africa.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Active Comparator (Standard daily oral PrEP, e.g., emtricitabine) **Blinding:** Double-blind (Standard design for active-controlled oral PrEP trials) **Randomization:** Randomized

**3.2 Planned Enrollment & Duration Target \$\$\$:** 4,580 participants **Number of Sites:** Multiple sites across sub-Saharan Africa **Estimated Study Start:** 10-Nov-2025 **Estimated Primary Completion:** 18-Oct-2027

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Is confirmed Human Immunodeficiency Virus (HIV)-uninfected based on negative HIV-1/HIV-2 test results. 2. Has been sexually active (2 vaginal intercourse encounters with cisgender male individual(s) within the last 3 months). 3. Was assigned female sex at birth and is cisgender. 4. Weighs  $\geq 35$  kg.

**4.2 Exclusion Criteria** 1. Has hypersensitivity or other contraindication to any component of the study interventions. 2. Has evidence of acute or chronic hepatitis B infection. 3. Has a history of malignancy within 5 years of screening except for adequately treated basal cell or squamous cell skin cancer, or in situ cervical cancer. 4. Has taken cabotegravir, lenacapavir, or any other long-acting HIV prevention product at any time. 5. Is receiving or is anticipated to require any prohibited therapies from 30 days prior to Day 1 through the study duration. 6. Has received an HIV vaccine at any time (ie, through past participation in an investigational clinical study) or monoclonal antibodies to HIV within 12 months before Day 1.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** MK-8527 administered once a month. **Active Comparator:** Standard daily oral PrEP (emtricitabine-based regimen). **Route of Administration:** Oral. **Duration of Treatment:** Up to study completion (approximately 96 weeks / 2 years).

**5.2 Concomitant & Prohibited Therapy Permitted:** Routine concomitant medications that do not exhibit significant drug-drug interactions with MK-8527. **Prohibited:** Concurrent use of other antiretroviral therapies, prohibited therapies from 30 days prior to Day 1, or investigational HIV PrEP agents outside of the study design.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                                   |
|--------------|---------------|---------------------------------------------------------------------------------------------|
| Screening    | Day -28 to 0  | Informed Consent, HIV-1/HIV-2 Testing, Medical History, Hepatitis B Testing, Pregnancy Test |
| Baseline     | Day 1         | Randomization, First Dose, Vitals, Baseline Efficacy & Safety Labs                          |
| Interim      | Monthly       | HIV Testing, Safety Labs, IP Administration/Dispensing, Adherence Counseling                |
| End of Study | Week ~96      | Final HIV Test, Exit Interview, IP Accountability, Full Safety Review                       |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Outcomes**

- 1. Number of Participants with Adjudicated Human Immunodeficiency Virus Type 1 (HIV-1) Infection:** The number of participants with adjudicated HIV-1 infection will be presented.
- 2. Number of Participants who Experience At Least One Adverse Event (AE):** An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. The number of participants that experienced AEs will be presented.
- 3. Number of Participants who Discontinue Study Intervention Due to an AE:** An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. The number of participants that discontinued study intervention due to an AE will be presented.

**7.2 Secondary & Exploratory Endpoints** \* Plasma and intracellular pharmacokinetics (PK) of MK-8527 and its active metabolites.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring of AEs via investigator assessment, participant interviews, and routine laboratory tests. An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent External Data Monitoring Committee (eDMC) will review unblinded safety and efficacy data at prespecified intervals.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) Set for the primary efficacy analysis (HIV-1 infection incidence). Safety Analysis Set for all patients who receive at least one dose of the study drug. **Sample Size Justification:** Target  $N = 4,580$  to provide adequate statistical power ( $\alpha = 0.05$ ) to establish non-inferiority or superiority of MK-8527 compared to the standard-of-care daily PrEP regimen regarding HIV incidence. **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier estimates) and multiple imputation for missing virologic data where appropriate.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

**Monitoring Plan:** Continuous site monitoring, incorporating both remote data review and onsite visits to ensure protocol and regulatory adherence.

**Audit Trail:** Fully validated 21 CFR Part 11 compliant Electronic Data Capture (EDC) system with full audit trails for eCRFs and data clarification forms.

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## 11. Study Identifiers & Governance

**Primary ID:** MK-8527-010 **Registry Number:** NCT07071623 **Sponsor/Lead Organization:** Merck **Study Title:** A Study of MK-8527 to Prevent Human Immunodeficiency Virus Type 1 (HIV-1) (MK-8527-010)

**Official Regulatory Title:** A Phase 3, Randomized, Active-Controlled, Double-Blind Clinical Study to Evaluate the Efficacy and Safety of MK-8527 Oral Once-Monthly as HIV-1 Preexposure Prophylaxis in Women

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## 12. Clinical Framework (HIV PrEP Specific)

**Primary Condition:** Human Immunodeficiency Virus (HIV) Pre-Exposure Prophylaxis (PrEP) **Diagnosis Threshold:** Confirmed HIV-negative status prior to enrollment **Medical Methodology:** Guideline-directed HIV Pre-Exposure Prophylaxis **Optimization**

**Target:** Sustained virologic protection (maintenance of HIV-uninfected status) combined with high patient adherence

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard PrEP clinical pathway (regular sexually transmitted infection screening, HIV testing, and risk-reduction counseling) **Education Protocol (HCP):** PrEP administration, medication adherence tracking, and HIV diagnostic algorithms **Education Protocol (Patient):** Safe sex practices, adherence to the monthly dosing schedule, and regular clinical monitoring **Quality Audit Mechanism:** Pill counts, observed dosing for the monthly administration, and participant adherence questionnaires

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## 14. Observation & Follow-Up Schedule

**Total Duration:** ~104 Follow-up weeks (Until October 2027) **Onsite**

**Visit Cadence:** Monthly visits (aligned to the once-monthly MK-8527 dosing regimen)

**Remote Monitoring Frequency:** Between clinic visits as clinically indicated

**Checklist Review Interval:** Monthly safety and efficacy checklist review

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15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|------|------|------|------|--|
| Role                     | Name   | Signature | Date          | :--- | :--- | :--- | :--- |  |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |  |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |      |      | Lead |      |  |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |  |